

Duality structures on tensor categories coming from vertex operator algebras

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Overview

- ① Vertex (operator) algebras and commutative algebras
- ② Tensor products
- ③ Duality structures

Definition: Vertex algebra (VA)

Data:

Axioms:

- $\mathbb{C}$ vector space V
- field map
 $Y_z : V \otimes V \rightarrow V[[z]]$
- vacuum vector $\Omega \in V$.
- vacuum axiom:
 $Y_z(\Omega \otimes -) = \text{id}_V$ and
 $Y_z(A \otimes \Omega) = A + zV[[z]], \forall A \in V$
- locality: $\forall A, B \in V, \exists n \in \mathbb{N}$
 $(z - w)^n [Y_z(A \otimes -), Y_w(B \otimes -)] = 0$

Consequence/Proposition

- There exists a translation operator $T : V \rightarrow V$,
such that $T\Omega = 0$ and $[T, Y_z] = \partial_z Y_z$
- For all $A, B, C \in V$

$$Y_z(A \otimes Y_w(B \otimes C)) \in V[[z]][[w]]$$

$$Y_w(B \otimes Y_z(A \otimes C)) \in V[[w]][[z]]$$

$$Y_w(Y_{z-w}(A \otimes B) \otimes C) \in V[[w]][[z - w]]$$

are expansions of the same element in $V[[z, w]][[z^{-1}, w^{-1}, (z - w)^{-1}]]$.

Example: Heisenberg vertex algebra

- Heisenberg Lie algebra: $\mathfrak{h} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C}a_n \oplus \mathbb{C}\mathbf{1}$ with $\mathbf{1}$ central and relations $[a_m, a_n] = m\delta_{m+n,0}\mathbf{1}$.
- Fock space ($\mathfrak{h}$ Verma module), $F_\lambda = \mathbb{C}[a_{-n} : n \geq 1] |\lambda\rangle$, where $a_0 = \lambda \text{id}$, $a_n/n = \partial_{a_{-n}}$, $n \geq 1$.
- The assignments and recursions

$$|0\rangle \otimes - \mapsto \text{id}_{F_0}, \quad a_{-1} |0\rangle \otimes - \mapsto \sum_{i \in \mathbb{Z}} a_i z^{-i-1} = a(z),$$

$$a_{-n} |0\rangle \otimes - \mapsto \frac{\partial^{n-1}}{(n-1)!} a(z) \otimes -,$$

$$a_{-n} v \otimes - \mapsto \left(\frac{\partial^{n-1}}{(n-1)!} a(z) \right)_r Y_z(v \otimes -) + Y_z \left(v \otimes \left(\frac{\partial^{n-1}}{(n-1)!} a(z) \right)_p \right)_-$$

define a vertex algebra on F_0 with vacuum vector $|0\rangle$.

Definition: Vertex operator algebra (VOA)/conformal vertex algebra

A vertex operator algebra is a vertex algebra (V, Y, Ω, T) admitting a conformal vector $\omega \in V$ satisfying

- Virasoro algebra relations: $Y_z(\omega \otimes -) = \sum_{n \in \mathbb{Z}} L_n z^{-n-2}$

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0} ,$$

with $c \in \mathbb{C}$, central charge.

- Non-negative integral conformal grading:

$$V = \bigoplus_{n=0}^{\infty} V_n, \quad V_n = \{v \in V \mid (L_0 - n)v = 0\}.$$

- Conformal derivation: $L_{-1} = T$.

Example: Heisenberg vertex operator algebra

Consider the vertex algebra (F_0, Y) from before.

- For $\rho \in \mathbb{C}$, the vector $\omega_\rho = (\frac{1}{2}a_{-1}^2 + \rho a_{-2}) |0\rangle \mapsto \frac{1}{2} (a(z)_r a(z) + a(z)a(z)_p) + \rho \partial a(z)$ is conformal.
- The central charge of the resulting Virasoro algebra is $c_\rho = 1 - 12\rho^2$.
- The conformal grading on F_0 assigns grade 0 to $|0\rangle$ and grade $-n$ to a_n .

Recap

- Vertex algebras are essentially associative commutative unital $\mathbb{C}$ -algebras with a derivation.
- Fields $Y_z : V \otimes V \rightarrow V((z))$ are essentially an action of V on itself.
- Where you have commutative algebras, you have modules!

Definition: Vertex algebra module

Let (V, Ω, T, Y) be a vertex algebra. A V -module is a pair (M, Y^M) : M a vector space and $Y_z^M : V \otimes M \rightarrow M((z))$ a V -action, that is,

- $Y_z^M(\Omega \otimes -) = \text{id}_M$
- For all $A, B \in V$ and $C \in M$ the expansions
$$Y_z^M(A \otimes Y_w^M(B \otimes C)) \in M((z))((w))$$
$$Y_w^M(B \otimes Y_z^M(A \otimes C)) \in M((w))((z))$$
$$Y_w^M(Y_{z-w}(A \otimes B) \otimes C) \in M((w))((z-w))$$
can be identified in $M[[z, w]][z^{-1}, w^{-1}, (z-w)^{-1}]$.

Many additional assumptions can be added to the above definition.
E.g. bounded conformal weights, finite weight spaces, semisimplicity, etc.

Example: Heisenberg modules

- Recall the Heisenberg algebra $\mathfrak{h}$ and Fock spaces F_λ from before.
- For $\lambda \in \mathbb{C}$, the Fock space F_λ is a (F_0, Y) -module with the action defined by the same formula, e.g.
$$Y_z^{F_\lambda}(a_{-1} |0\rangle \otimes |\lambda\rangle) = a(z) |\lambda\rangle.$$

Duals of modules

Definition: the dual of a module

Let (V, Y, Ω, ω) be a vertex operator algebra and (M, Y^M) , $M = \bigoplus_n M_n$ a module. Then $(M', Y^{M'})$

- $M' = \bigoplus_n \text{Hom}_{\mathbb{C}}(M_n, \mathbb{C})$,
- $\langle Y_z^{M'}(v \otimes \mu), m \rangle = \langle \mu, Y_{z^{-1}}^M(e^{zL_1}(-z^2)^{L_0}v \otimes m) \rangle$, $v \in V, m \in M, \mu \in M'$.

is again a (V, Y) -module.

Heisenberg example:

For the choice of conformal vector ω_ρ , we have $F'_\lambda \cong F_{2\rho-\lambda}$

Motivating tensor products

- In quantum field theory all information is encoded in n -point correlation functions.
- In conformal quantum field theory (CFT) these correlation functions are V -multilinear functions.
- So we need to understand multilinear algebra for vertex algebras.
- This is also a natural question for commutative algebras.

Definition: Intertwining operator, V -bilinear maps

Let (V, Ω, Y, ω) be a vertex operator algebra and $(M_1, Y^{M_1}), (M_2, Y^{M_2}), (M_3, Y^{M_3})$ be V -modules. An intertwining operator of type $\binom{M_3}{M_1, M_2}$ is a map $\mathcal{Y}_x : M_1 \otimes M_2 \rightarrow M_3\{x\}$ such that for all $m_i \in M_i$

- $\mathcal{Y}_x(m_1 \otimes m_2)$ truncates below.
- $\mathcal{Y}_x(L_{-1}m_1 \otimes m_2) = \partial_x \mathcal{Y}_x(m_1 \otimes m_2)$.
- The expansions $Y_z^{M_3}(A \otimes \mathcal{Y}_x(m_1 \otimes m_2)) \sim \mathcal{Y}_x(Y_{z-x}^{M_1}(A \otimes m_1) \otimes m_2) \sim \mathcal{Y}_x(m_1 \otimes Y_z^{M_2}(A \otimes m_2))$ can be identified.

Observations:

- The field map Y is an intertwining operator of type $\binom{V}{V, V}$.
- The action Y^M is an intertwining operator of type $\binom{M}{V, M}$.
- Intertwining operators are V -bilinear maps. All intertwining operators of a given type form a vector space. The field map Y and the action Y^M have a distinguished normalisation due to $Y_z(\Omega \otimes -) = \text{id}$.

Example: Heisenberg intertwining operators

Recall the Heisenberg Fock spaces F_μ , $\mu \in \mathbb{C}$.

Then $\dim ({}^{F_\rho}_{F_\mu, F_\nu}) = \delta_{\rho, \mu+\nu}$ for all $\rho, \mu, \nu \in \mathbb{C}$.

$({}^{F_{\mu+\nu}}_{F_\mu, F_\nu})$ is spanned by

$$\mathcal{Y}_x^{F_\mu, F_\nu}(p|\mu\rangle \otimes q|\nu\rangle) = x^{\mu\nu} S_\mu \prod_{m \geq 1} \exp\left(\mu \frac{a_{-m}}{m} x^m\right) Y_x^{F_\nu}(p|0\rangle \otimes -) \\ \cdot \prod_{m \geq 1} \exp\left(-\mu \frac{a_m}{m} x^{-m}\right) q|\nu\rangle,$$

where $S_\mu : |\nu\rangle \mapsto |\mu + \nu\rangle$ is the shift operator.

Tensor products pull multilinear algebra back to linear algebra!

Definition: Fusion product aka vertex algebra tensor product

Let (V, Ω, Y, ω) be a vertex operator algebra and $(M_1, Y^{M_1}), (M_2, Y^{M_2})$ be V -modules. A fusion product is a triple $(M_1 \boxtimes M_2, Y^{M_1 \boxtimes M_2}, \mathcal{Y}^{M_1, M_2})$, where $(M_1 \boxtimes M_2, Y^{M_1 \boxtimes M_2})$ is a V -module and $\mathcal{Y}^{M_1, M_2}$ is an intertwining operator of type $\begin{pmatrix} M_1 \boxtimes M_2 \\ M_1, M_2 \end{pmatrix}$ such that the following universal property holds: For every V -module (X, Y^X) and intertwining operator $\mathcal{Y}^X$ of type $\begin{pmatrix} X \\ M_1, M_2 \end{pmatrix}$

$$\begin{array}{ccc}
 M_1 \otimes M_2 & \xrightarrow{\mathcal{Y}^{M_1, M_2}} & M_1 \boxtimes M_2 \{z\} \\
 & \searrow \mathcal{Y}^X & \downarrow \exists! f \\
 & & X \{z\}
 \end{array}$$

In contrast to linear algebra (or ring theory) constructing $M_1 \boxtimes M_2$ and decomposing into a direct sum of indecomposable modules is **extremely** hard.

Well chosen categories of modules are tensor categories with respect to $\boxtimes$ with the following structures. [Huang-Lepowsky-Zhang]

- For module homomorphisms $f : X \rightarrow Z$, $g : U \rightarrow W$, the morphism $f \boxtimes g$ is uniquely characterised by

$$(f \boxtimes g) \circ \mathcal{Y}^{X \boxtimes U} = \mathcal{Y}^{Z \boxtimes W} \circ (f \otimes g)$$

- V is the tensor identity and the unit isomorphisms are uniquely characterised by

$$\ell_M \left(\mathcal{Y}_z^{V, M}(a \otimes m) \right) = Y_z^M(a \otimes m) \text{ and}$$

$$r_M \left(\mathcal{Y}_z^{M, V}(m \otimes a) \right) = e^{zL-1} Y_{-z}^M(a \otimes m).$$

- Associativity isomorphisms (hardest part!)

$$A_{M_1, M_2, M_3} \left(\mathcal{Y}_{x_1}^{M_1, M_2 \boxtimes M_3}(m_1 \otimes \mathcal{Y}_{x_2}^{M_2, M_3}(m_2 \otimes m_3)) \right) =$$

$$\mathcal{Y}_{x_2}^{M_1 \boxtimes M_2, M_3}(\mathcal{Y}_{x_1 - x_2}^{M_1, M_2}(m_1 \otimes m_2) \otimes m_3)$$

All analytic details hidden.

- Braiding isomorphisms uniquely characterised by

$$c_{M_1, M_2} \left(\mathcal{Y}_x^{M_1, M_2}(m_1 \otimes m_2) \right) = e^{xL-1} \mathcal{Y}_{e^{i\pi}x}^{M_2, M_1}(m_2 \otimes m_1)$$

- If the vertex algebra V is conformal (a vertex operator algebra) and the modules are chosen to be compatible with this conformal structure, then there is also a twist $\theta_M = e^{2\pi i L_0}|_M$, which satisfies the balancing equation

$$\theta_{M_1 \boxtimes M_2} = c_{M_1, M_2} \circ c_{M_2, M_1} \circ (\theta_{M_1} \boxtimes \theta_{M_2})$$

- Tensor categories of vertex operator algebra modules depend only **very weakly** on the conformal structure. Only the twist and taking duals depend on the conformal structure.

Theorem [Huang, Moore-Seiberg]: The Verlinde conjecture

Let (V, Y, Ω, ω) be a vertex operator algebra and $\text{Adm}(V)$ be the category of admissible V -modules. If

- 1 $\dim V_0 = 1, \dim V_{-n} = 0, \dim V_n < \infty, n \in \mathbb{N}$,
- 2 V is simple as a module over itself,
- 3 $V \cong V'$, self-dual,
- 4 $\dim V/c_2(V) < \infty$, (a technical finiteness condition)
- 5 $\text{Adm}(V)$ is semisimple,

then $\text{Adm}(V)$ is a modular tensor category. Further the action of the modular group on the category (which determines Verlinde's formula) is equal (after a renormalisation) to the action of the modular group on module characters.

Recap

- Vertex algebras are almost commutative unital algebras with derivations.
- The conformal vector is a choice/structure: there can be 0, 1 or many.
- Vertex algebras admit modules. “Good choices” of module categories admit a tensor (aka fusion) product.
- With the exception of associators, the tensor structure morphisms follow from easy constructions.

Beyond modular tensor categories

The Verlinde conjecture is about vertex operator algebras that are *maximally nice*. There are many deviations from niceness.

- The category of modules need not be finite (the Heisenberg example is not finite).
- The category of modules need not be semisimple.
- The tensor product need not be left exact (there can be non-flat objects). [Gaberdiel-Runkel-SW]
- The first two deviations above can still be nice in the sense that they admit rigid duals. [Tsuchiya-SW, Allen-SW]
The third deviation is more fundamentally broken, as non-flat objects cannot have rigid duals.

Duals of intertwining operators

Proposition [HLZ]

Let $\mathcal{Y}$ be an intertwining operator of type $\left(\begin{smallmatrix} M_3 \\ M_1, M_2 \end{smallmatrix}\right)$. Then

$$\begin{aligned} M_1 \otimes M'_3 &\rightarrow M'_2\{x\} \\ m \otimes \nu &\mapsto \nu(\mathcal{Y}_{x^{-1}}(e^{xL_1}(-x^2)^{L_0}m \otimes -)) \end{aligned}$$

is an intertwining operator of type $\left(\begin{smallmatrix} M'_2 \\ M_1, M'_3 \end{smallmatrix}\right)$.

Thus $\left(\begin{smallmatrix} M_3 \\ M_1, M_2 \end{smallmatrix}\right) \cong \left(\begin{smallmatrix} M'_2 \\ M_1, M'_3 \end{smallmatrix}\right)$.

Remark

Intertwining operators of type $\left(\begin{smallmatrix} M_3 \\ M_1, M_2 \end{smallmatrix}\right)$ should be thought of as Hom -spaces of the form $\text{Hom}_V(M_1 \boxtimes M_2, M_3)$.

Definition: Grothendieck-Verdier categories

Let $\mathcal{C}$ be a monoidal (abelian linear) category.

- 1 An object $K \in \mathcal{C}$ is called *dualising*, if for all $Y \in \mathcal{C}$ the functor $\mathrm{Hom}_{\mathcal{C}}(- \otimes Y, K)$ is representable, that is,

$$\mathrm{Hom}_{\mathcal{C}}(X \otimes Y, K) \cong \mathrm{Hom}_{\mathcal{C}}(X, GY),$$

and if the so defined contravariant functor $G : \mathcal{C} \rightarrow \mathcal{C}$ is an anti-equivalence.

- 2 A monoidal category $\mathcal{C}$ together with a choice of dualising object $K \in \mathcal{C}$ is called a *Grothendieck-Verdier* or **-autonomous* category.

Theorem [Allen-Lentner-Schweigert-SW]

Let (V, Y) be a vertex operator algebra and let $\mathrm{Rep}(V)$ be a choice of modules to which the HLZ tensor product theory applies which is in addition closed under taking duals. Then V' is a dualising object for $\mathrm{Rep}(V)$ and $(\mathrm{Rep}(V), V')$ is a Grothendieck-Verdier category (actually ribbon Grothendieck-Verdier).

Why am I excited about this?

Grothendieck-Verdier categories have many appealing features

- They appear to describe the *natural* duality structure of vertex operator algebra modules and they can accommodate all non-nice features mentioned previously.
- They admit two tensor products $X \otimes Y$ and $X \bullet Y = G(G^{-1}Y \otimes G^{-1}X)$, where $\otimes$ is right exact and $\bullet$ is left exact. In particular there are distributor morphisms

$$\partial_{X,Y,Z}^l : X \otimes (Y \bullet Z) \rightarrow (X \otimes Y) \bullet Z, \quad \partial_{X,Y,Z}^r : (X \bullet Y) \otimes Z \rightarrow X \bullet (Y \otimes Z),$$

which need not be isomorphisms. [Fuchs-Schaumann-Schweigert-SW]

- $\otimes$ admits inner Homs (right adjoints) and $\bullet$ admits inner coHoms (left adjoints), which allow the construction of algebras and coalgebras (even Frobenius algebras) in $\mathcal{C}$ (crucial for conformal field theory).

Thank you!

For further reading I



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